



Semester I Examinations 2017/2018

Exam Code(s)	1SD1/1SD3/1MF1
Exam(s)	Higher Diploma in Applied Science (Software Design & Development), Higher Diploma in Applied Science (Software Design & Development) Industry Stream, M.Sc. in Software Design & Development
Module Code(s)	CT874
Module(s)	Programming I
Paper No.	I
External Examiner(s)	Professor Pier Luca Lanzi
Internal Examiner(s)	Dr. Michael Schukat * Seamus Hill
<u>Instructions:</u>	Candidates are required to answer: Section A (20 Marks) Answer all 10 parts of section A, This section consists of 10 multiple-choice questions all of which should be attempted. Answers are to be written on the MCQ sheet provided, NOT on the Examination paper. Section B (80 Marks) Answer Any Two Questions.
Duration	2 hours
No. of Pages	7
Discipline(s)	Information Technology
Requirements	MCQ

Section A (20 Marks)
Attempt all 10 questions
Mark your answers on the MCQ sheet provided

- 1) Polymorphism allows us to _____.
- a) have the same message execute a different method depending on the receiving object.
 - b) hide the internal workings of a Java program from the client programmer.
 - c) overload methods and allow the same method name to be used multiple times.
 - d) extend and modify code.
- 2) Assume we have three classes, X Y and Z. Y and Z are siblings and are subclasses of the X class. Which classes have access to a private variable in class X?
- a) X, Y and Z
 - b) X and Y only
 - c) Y and Z only
 - d) X only
- 3) How would we access the 12th element of the array, x?
- a) x(12);
 - b) x{12};
 - c) x[11];
 - d) x[12];
- 4) Which of the following methods is not contained within the List class?
- a) hasNext()
 - b) add()
 - c) List()
 - d) remove()
- 5) When an array is passed to a method _____.
- a) only the reference is passed, and a copy of the array is not created in the method.
 - b) a copy of the array is created within the method.
 - c) the value of the array is passed into the method.
 - d) None of the above.

- 6) The practice of writing multiple constructors to handle different sets of inputs is known as _____.
- a) Inheritance
 - b) Polyinputting
 - c) Overloading
 - d) Multiplicity
- 7) How could a programmer stick to common practice with respect to initializing class variables through constructors?
- a) Use the same name for the parameter corresponding to the class variable as for the class variable.
 - b) Use a different name for the parameter corresponding to the class variable as for the class variable.
 - c) Always include multiple constructors.
 - d) Use the keyword, this, as much as possible.
- 8) What is meant by “returning an object” from a method?
- a) It means you are returning a null value to the caller.
 - b) It means returning a primitive type to the caller.
 - c) It means returning the object’s address to the caller.
 - d) It means returning the object’s value to the caller.
- 9) What is the difference between a sentinel-controlled loop and a count-controlled loop?
- a) A count-controlled loop is executed for a fixed number of times while a sentinel-controlled loop is executed infinitely.
 - b) A count-controlled loop uses the numerical increment operator ++, while the sentinel-controlled loop uses Boolean expressions.
 - c) A count-controlled loop is executed for a fixed number of times while a sentinel-controlled loop is executed repeatedly until a certain value is encountered.
 - d) A count-controlled loop does not take user input in its body, while a sentinel-controlled loop does.
- 10) In computer science, what is meant by garbage collection?
- a) Deallocation of unused memory space.
 - b) Going through the program and fixing erroneous code.
 - c) Deletion of objects from memory.
 - d) Defragmentation of the disk drive.

Section B (80 Marks)
Answer Any TWO Questions

Q 2.

- a) Write an application that declares, creates and randomly initialises a *LinkedList* of 10 *Integer* objects, using an *enhanced for loop* to display the contents.

[10]

- b) The application should include a method that writes the contents of the *LinkedList* to a text file. The method is to have the following signature:

```
public static void writeToFile(List<Integer> list)
```

[15]

- c) The application should also include a method that reads and displays the contents from the text file. The method signature is as follows:

```
public static void readFromFile(String filename)
```

[15]

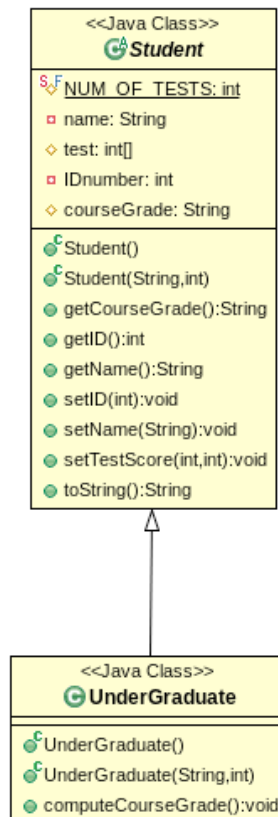
PTO

Q 3.

- a) In relation to the visibility of data members, discuss the effects of the modifiers *protected*, *public* and *private*, using examples where appropriate.

[10]

- b) From the UML diagram below you are required to:



- i. Implement the class hierarchy as illustrated.

Note: `computeCourseGrade()` calculates the average mark for each student, based on the test scores stored in the `test` array and sets the `courseGrade` field to *Pass* or *Fail*.

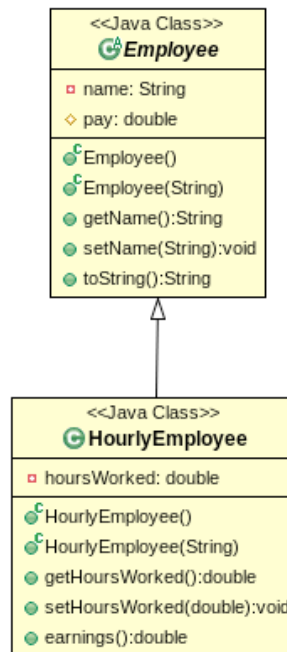
[15]

- ii. Write a driver class that creates an array of type *Student* and populate it with 3 *UnderGraduate* instances. The driver class should allow the user enter individual test scores, which are stored in the `test` array, for each test taken by a student and display the student name, ID number and whether or not they have passed or failed.

[15]

PTO

Q 4. From the UML diagram below you are required to:



- a) Implement the classes outlined in the UML diagram above, ensuring the classes are *serializable*.

Note:

- Employees are paid a rate of €15 per hour.
- The Employee class contains the following abstract method:
abstract double earnings();

[15 Marks]

- b) Write a driver class to carry out the following:

- Create an *ArrayList* of employees and populate the list with two instances of the subclass.
- Display the employees name with their calculated earnings using *toString()*.
- Write the output to a file called *employee.txt*.
- Serialize the List.

[25 Marks]

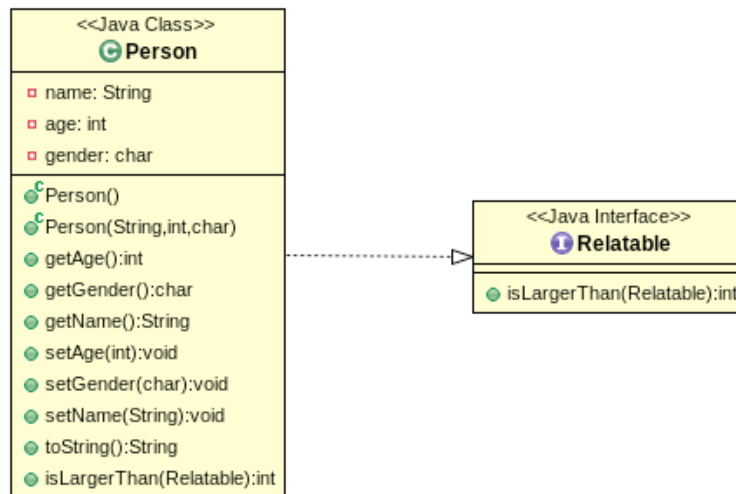
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Q 5.

- a) In relation to Java describe in your own words, what an *Interface* is.

[5]

- b) Using the UML diagram outlined, write a Java application that implements the **Person** class and **Relatable** interface to allow comparison between *Person* objects based on *age*.



Note: *isLargerThan()* returns a 1, 0 or -1 value depending on whether an object is greater than, equal to or less than a similar type of object.

[15]

- c) Write a driver class which:

- i. Creates and populates an *ArrayList* with three instances of the *Person* class.

[6]

- ii. Includes a method with the following signature:

public static Person findOldest(List<Person> p)

The method uses a *for loop* to traverse the List calling **isLargerThan** to determine the oldest person in the List, which is then returned.

[12]

- iii. Displays the oldest person in the List using *toString()*.

[2]